Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

PENDAHULUAN

TINJAUAN PUSTAKA

[utf8]inputenc lmodern [T1]fontenc [english]babel textcomp amsmath mathrsfs latexsym amssymb amsfonts theorem graphicx scrlayer-scrpage xcolor setspace framed hyperref pgf,tikz,pgfplots mathrsfs compat=1.15arrows qrcode multicol multirow xurl tabularx enumitem float colortbl,rotating,booktabs

grogray0.6 gro Athénée de Luxembourg gro Write your title here gro Your name gro gro - **1** - gro

Write your title here

Write your research question here

Your name

Athénée de Luxembourg

Mathematics - your class

Investigation/IA/EE

Word Count: xx

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1 Introduction

With the purpose of discovering the different fields of study within the umbrella term mathematics, I came across numerical analysis which sparked my interest as it implements algorithms with the aim of solving math- ematical analysis problems with continuous variables. After some further research, I discovered the Weierstrass approximation theorem, defining the key element of polynomial approximation being that *'any continuous function on a closed and bounded interval can be uniformly approximated on that interval by polynomials to any degree of accuracy.'* [1] Essentially, this states that a graph which consists of a single unbroken curve in a defined domain can be expressed by a polynomial function whose precision in approximating is determined by its degree. With the aim of analyzing the application of this theorem, I will consider a continuous function which I will approximate with three different polynomial approximation methods.

2 A few lines of calculations

The formula of a slope can be proven as follows. For the points $A(x_1; y_1)$ and $B(x_2; y_2)$, that satisfy the equation, I establish:

$$\{ y_1 = mx_1 + b$$

$y_2 = mx_2 + b$ Thus, I can subtract one of these from the other:

$$\begin{aligned} & y_2 - y_1 && = mx_1 + b - (mx_2 + b) \\ \Leftrightarrow\ & y_2 - y_1 && = mx_1 - mx_2 \\ \Leftrightarrow\ & m && = y_2 - y_1 x_2 - x_1 \end{aligned}$$

This can be rewritten to find the formula, I used, of a straight line:

$$y_2 \quad = \quad y_1 + y_2 - y_1 x_2 - x_1 \cdot (x_1 - x_2)$$

3 A matrix

To continue, we transfer our values into an augmented matrix.

$$714316$$

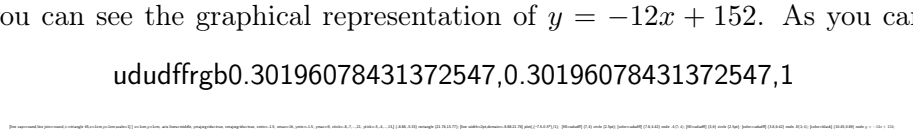
The next step is to add (-3) times the first line to 7 times the second line to triangulize the matrix.

$$\Leftrightarrow 21312 red 0 - 4 - 30$$

[1] https://math.stackexchange.com/questions/tagged/weierstrass-approximation

4 Inserting a graph from GeoGebra

Below you can see the graphical representation of $y = -12x + 152$. As you can see in



Gambar 1: Linear model representing $y = -\frac{1}{2}x + \frac{15}{2}$

Figure 1, the line passes through the two previously chosen points; $A(7; 4)$ and $B(3; 6)$, which indicates that our model is correct, that we made no mistakes in our calculations.

5 Creating a table

Fill in the following divisibility table.

	2	3	4	5	6	7	9	11	15
3 465									
4 725									

Tabel 1: Divisibility table

6 Inserting QR Codes

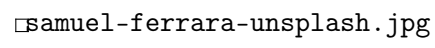
2

How to install and use the Excel add in *Excel2Latex*

https://express.adobe.com/video/1Dx733VtuxUw9
YouTube video: Secant Method | Oscar Veliz

https://youtu.be/$_{M}fjXOLUnyw?t = 65$

7 Inserting an image



Gambar 2: Weisshorn, Randa, Switzerland. Author: Samuel Ferrara

8 Customizing the enumerate environment

2

[label=.]Typesetting[label=0)]

1. a) Written in LaTeX and includes
   b) Cover page
   c) Header/footer
   d) Table of contents
   e) Introduction and conclusion
2. Format

   [label=)]Divide into different sections/subsections Each mathematical expression inside math mode ($ $ or $$ $$) Each figure should have a caption Reference figures correctly in your text

9 Customizing the itemize environment

- a) First draft: 31-Feb-2029 9pm via MS Teams Assignment
- ∘ Final report: 31-Sep-2029 9pm via MS Teams Assignment
- ∗ Submit as a PDF file via Teams
- – Name of draft: Name-FirstName-Draft.pdf

  Name of final report: Name-FirstName-Final.pdf

10 Bibliography

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